

GIRIDHAR S

📍 Chennai, India | ✉ giridharsridharan11@gmail.com | ☎ (+91) 915-031-0513 | in Giridhar S | 🌐 GIRIDHAR-VLSI-Engineer

Profile

Final-year Electronics Engineering student specialising in VLSI Design and Embedded Systems, with hands-on experience in physical design automation, GDSII layout, and hardware prototyping. Proven through a state-level hackathon prototype and a VLSI IP-protection tool built with Cadence Virtuoso and KLayout. Actively seeking internship or full-time roles in VLSI/RTL Design or Embedded Systems engineering.

Education

S.A. Engineering College B.E. Electronics Engineering — VLSI Design and Technology	Chennai, India 2023 – 2027
St. Joseph's Matriculation Higher Secondary School Higher Secondary Certificate (HSC)	Chennai, India 2023
St. Joseph's Matriculation Higher Secondary School Secondary School Leaving Certificate (SSLC)	Chennai, India 2021

Technical Skills

Languages	Python, Verilog HDL, C++
EDA Tools	KLayout 0.28, Cadence Virtuoso 6.1.7, Assura DRC, Xilinx Vivado, Proteus 8 Professional
VLSI	GDSII Layout, Physical Design, DRC, Dummy Cell Insertion, IP Protection
Libraries	gdsapy, pya (KLayout API), Streamlit, NumPy, Pandas, scikit-learn, librosa, Flask
Embedded	Arduino (Uno R3, Nano), ESP32, HC-12 Wireless Communication, RF Communication, Ultrasonic Sensors, LCD, Servo & Stepper Motors, INMP441 I2S Mic
Other Tools	MATLAB, PCB Design, React (basics), VMware Workstation, Ubuntu 24.04 LTS

Projects

Python-Based Layout-Level Dummy Cell Watermarking for IP Protection

Tools: Python, KLayout pya API, Cadence Virtuoso, Assura DRC, GPDk045, Streamlit

- Built an automated VLSI IP-protection tool embedding hidden owner identity as scattered dummy cells in GDSII layouts using BASE64 encoding and SHA-256 cryptographic verification.
- Verified watermark detection on CMOS inverter layouts with zero DRC violations and zero power impact using Cadence Virtuoso + GPDk045 45nm PDK.
- Developed a Streamlit web app supporting real GDS processing, attack simulation, and SHA-256 ownership verification.

Smart Curve Alert System (SCAS) — Hardware Prototype

TN-IMPACT 2026 Hackathon, TANCAME (Govt. of Tamil Nadu) | Team: ASTRIX

- Designed and built a two-module embedded system using Arduino, HC-12 wireless modules, and ultrasonic sensors to detect vehicles at blind curves and deliver real-time alerts.
- Implemented a solar-powered roadside transmitter pole with LED alerts and an in-vehicle receiver with LCD and voice output.
- Presented working physical prototype at **TN-IMPACT 2026**, a state-level hackathon conducted by TANCAME, Government of Tamil Nadu.
- Proposed future integration with EV CAN Bus (ADAS) and V2X/Bluetooth for smart city deployment.

Anti-Deepfake Audio Authentication System

Tools: ESP32, INMP441 I2S Mic, Flask, Python, scikit-learn, librosa, React

- Building a hardware-software system embedding inaudible ultrasonic watermarks at the point of recording using ESP32 with INMP441 I2S microphone.

- Developing Flask/Python backend with cryptographic signing and a scikit-learn/librosa ML model for AI voice detection.
- Integrating React frontend for real-time verification of audio authenticity.

Internships & Training

Intern — 4SG Technology

2 months

Hardware Servicing — Canon, Brother, Epson Copiers

- Serviced and repaired 3–4 Canon, Brother, and Epson copiers daily — covering roller drives, fuser units, toner mechanisms, and gear assemblies.
- Diagnosed and resolved paper feed errors, alignment faults, and mechanical failures, completing most cases on-site without escalation.
- Gained working knowledge of laser and inkjet printing technologies through hands-on repair and preventive maintenance.
- Awarded certificate and Letter of Recommendation for technical competence and professional conduct.

Intern — MaestroMinds — Associate (Project-Based)

30-day Internship + Ongoing

Hardware Projects — Remote Collaboration (Zoho Cliq)

- Contributed to 2 real-time client hardware projects during the internship, working within professional delivery timelines and team workflows.
- Gained hands-on experience with remote collaboration and client communication tools (Zoho Cliq) in a professional project environment.
- Retained as a project-based associate post-internship — assigned to new hardware projects as they arise, with stipend on successful delivery, reflecting consistent performance and trust from the organisation.

Trainee — MSME-DFO Guindy, Govt. of India

30 days

Entrepreneurship Skill Development Programme (ESDP) — IoT Embedded Systems

- Built hands-on proficiency with Arduino Uno R3, Nano, and ESP32 — covering microcontroller architecture, programming, and real-world use cases.
- Worked with core embedded peripherals including LCD displays, servo motors, stepper motors, and potentiometer-based motor control circuits.
- Applied knowledge gained here directly in the Smart Curve Alert System (SCAS) hardware prototype, demonstrating practical continuity from training to project execution.
- Developed foundational understanding of business plan preparation and government MSME schemes for technology entrepreneurs.

Inplant Trainee — Chennai Port Authority

30 days

M&E Engineering Department — Electronics & Communication focus

- Completed inplant training across Electronic Data Processing, Electrical, Mechanical, and Electronics divisions, with primary focus on Electronics and Communication systems.
- Observed and studied large-scale industrial communication and electronic systems in an active port infrastructure environment — exposure not accessible in academic settings.
- Adhered to high standards of attendance, conduct, and work performance throughout the training period.

Printing Technician & Counter Staff — Printzo World

Since 2023, part-time ongoing

Print Operations, Customer Handling

- Independently managed print operations serving 150+ customers on peak days, often as the sole operator.
- Operated Canon, Brother, and Epson copiers for bulk and binding orders over 2+ years alongside full-time academics.

Certifications & Workshops

1. Python Masterclass — MRL Tech Solutions Pvt. Ltd.
2. PCB Design (Hands-on) — Pantech eLearning Pvt. Ltd.
3. Advanced Verilog HDL — Pantech eLearning Pvt. Ltd.
4. VLSI Design Flow (RTL to GDS) Workshop — ChipXpert Technologies Pvt. Ltd.